Year 11 Foundation

REVISION PACK

Progress Check 1

ANSWERS

Units in this pack

- Number 8
- Algebra 7

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 8 · Standard form (Lesson 1.1)

Q	Working	Answer
writing numbers in standard form		
1.1	**(a)** $2.7 \times 10\,000 = 27\,000$ **(b)** 6.3 6.3×10^{-3}	**(a) 27 000** **(b)** 6.3×10^{-3}
1.2	**(a)** $5.04 \times 100\,000 = 504\,000$ **(b)** 3.9 3.9×10^{-2}	**(a) 504 000** **(b)** 3.9×10^{-2}
1.3	**(a)** $9.1 \times 1000 = 9100$ **(b)** 8.05 8.05×10^{-3}	**(a) 9100** **(b)** 8.05×10^{-3}
1.4	**(a)** $6.25 \times 1000 = 6250$ **(b)** 5.1 5.1×10^{-4}	**(a) 6250** **(b)** 5.1×10^{-4}
1.5	**(a)** $1.8 \times 100\,000 = 180\,000$ **(b)** 2.7 2.7×10^{-3}	**(a) 180 000** **(b)** 2.7×10^{-3}
1.6	**(a)** $7.3 \times 1\,000\,000 = 7\,300\,000$ **(b)** 6.8 6.8×10^{-2}	**(a) 7 300 000** **(b)** 6.8×10^{-2}
multiplying in standard form		
1.7	$2 \times 3 = 6$ $10^{3+5} = 10^{8}$	6×10^{8}
1.8	$3 \times 3 = 9$ $10^{6+2} = 10^{8}$	9×10^{8}

1.9	$1.5 \times 2 = 3$ $10^{4+3} = 10^7$	3×10^7
1.10	$4 \times 2 = 8$ $10^{2+7} = 10^9$	8×10^9
1.11	$2.5 \times 3 = 7.5$ $10^{3+4} = 10^7$	7.5×10^7
1.12	$1.2 \times 4 = 4.8$ $10^{5+6} = 10^{11}$	4.8×10^{11}
big and small numbers		
1.13	**(a)** 3.8 3.8×10^6 **(b)** $2.6 \div 100 = 0.026$	**(a)** 3.8×10^6 **(b) 0.026**
1.14	**(a)** 6.5 6.5×10^4 **(b)** $7.1 \div 1000 = 0.0071$	**(a)** 6.5×10^4 **(b) 0.0071**
1.15	**(a)** 9.2 9.2×10^8 **(b)** $5.08 \div 10\,000 = 0.000508$	**(a)** 9.2×10^8 **(b) 0.000508**
1.16	**(a)** 4.1 4.1×10^5 **(b)** $1.9 \div 100 = 0.019$	**(a)** 4.1×10^5 **(b) 0.019**
1.17	**(a)** 2.8 2.8×10^7 **(b)** $6.45 \div 1000 = 0.00645$	**(a)** 2.8×10^7 **(b) 0.00645**
1.18	**(a)** 5.3 5.3×10^3 **(b)** $8 \div 100\,000 = 0.00008$	**(a)** 5.3×10^3 **(b) 0.00008**

Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

Q	Working	Answer
prime factors		
2.1	$90 = 2 \times 45$ $45 = 3 \times 15$ $15 = 3 \times 5$	$2 \times 3 \times 3 \times 5$
2.2	$126 = 2 \times 63$ $63 = 3 \times 21$ $21 = 3 \times 7$	$2 \times 3 \times 3 \times 7$
2.3	$150 = 2 \times 75$ $75 = 3 \times 25$ $25 = 5 \times 5$	$2 \times 3 \times 5 \times 5$
2.4	$48 = 2 \times 24$ $24 = 2 \times 12$ $12 = 2 \times 6$ $6 = 2 \times 3$	$2 \times 2 \times 2 \times 2 \times 3$
2.5	$200 = 2 \times 100$ $100 = 2 \times 50$ $50 = 2 \times 25$ $25 = 5 \times 5$	$2 \times 2 \times 2 \times 5 \times 5$
2.6	$132 = 2 \times 66$ $66 = 2 \times 33$ $33 = 3 \times 11$	$2 \times 2 \times 3 \times 11$
HCF and LCM		
2.7	**(a)** $20 : 1, 2, 4, 5, 10, 20$ $30 : 1, 2, 3, 5, 6, 10, 15, 30$ **(b)** $4 : 4, 8, 12$ $6 : 6, 12$	**(a) 10** **(b) 12**
2.8	**(a)** $16 : 1, 2, 4, 8, 16$ $40 : 1, 2, 4, 5, 8, 10, 20, 40$ **(b)** $6 : 6, 12, 18$ $9 : 9, 18$	**(a) 8** **(b) 18**

2.9	**(a)** $27 : 1, 3, 9, 27$ $45 : 1, 3, 5, 9, 15, 45$ **(b)** $8 : 8, 16, 24$ $12 : 12, 24$	**(a) 9** **(b) 24**
2.10	**(a)** $28 : 1, 2, 4, 7, 14, 28$ $42 : 1, 2, 3, 6, 7, 14, 21, 42$ **(b)** $5 : 5, 10, 15, 20, 25, 30, 35$ $7 : 7, 14, 21, 28, 35$	**(a) 14** **(b) 35**
2.11	**(a)** $32 : 1, 2, 4, 8, 16, 32$ $48 : 1, 2, 3, 4, 6, 8, 12, 16, 24, 48$ **(b)** $9 : 9, 18, 27, 36$ $12 : 12, 24, 36$	**(a) 16** **(b) 36**
2.12	**(a)** $15 : 1, 3, 5, 15$ $35 : 1, 5, 7, 35$ **(b)** $10 : 10, 20, 30$ $15 : 15, 30$	**(a) 5** **(b) 30**
prime factors and the HCF		
2.13	**(a)** $72 = 2 \times 36$ $36 = 2 \times 18$ $18 = 2 \times 9$ $9 = 3 \times 3$ **(b)** $72 = 2 \times 2 \times 2 \times 3 \times 3$ $54 = 2 \times 3 \times 3 \times 3$ $\text{HCF} = 2 \times 3 \times 3 = 18$	**(a)** $2 \times 2 \times 2 \times 3 \times 3$ **(b) 18**
2.14	**(a)** $120 = 2 \times 60$ $60 = 2 \times 30$ $30 = 2 \times 15$ $15 = 3 \times 5$ **(b)** $120 = 2 \times 2 \times 2 \times 3 \times 5$ $90 = 2 \times 3 \times 3 \times 5$ $\text{HCF} = 2 \times 3 \times 5 = 30$	**(a)** $2 \times 2 \times 2 \times 3 \times 5$ **(b) 30**

2.15	**(a)** $126 = 2 \times 63$ $63 = 3 \times 21$ $21 = 3 \times 7$ **(b)** $126 = 2 \times 3 \times 3 \times 7$ $84 = 2 \times 2 \times 3 \times 7$ $\text{HCF} = 2 \times 3 \times 7 = 42$	**(a)** $2 \times 3 \times 3 \times 7$ **(b) 42**
2.16	**(a)** $100 = 2 \times 50$ $50 = 2 \times 25$ $25 = 5 \times 5$ **(b)** $100 = 2 \times 2 \times 5 \times 5$ $75 = 3 \times 5 \times 5$ $\text{HCF} = 5 \times 5 = 25$	**(a)** $2 \times 2 \times 5 \times 5$ **(b) 25**
2.17	**(a)** $60 = 2 \times 30$ $30 = 2 \times 15$ $15 = 3 \times 5$ **(b)** $60 = 2 \times 2 \times 3 \times 5$ $45 = 3 \times 3 \times 5$ $\text{HCF} = 3 \times 5 = 15$	**(a)** $2 \times 2 \times 3 \times 5$ **(b) 15**
2.18	**(a)** $98 = 2 \times 49$ $49 = 7 \times 7$ **(b)** $98 = 2 \times 7 \times 7$ $70 = 2 \times 5 \times 7$ $\text{HCF} = 2 \times 7 = 14$	**(a)** $2 \times 7 \times 7$ **(b) 14**

Number 8 · Ratio (Lesson 1.4)

Q	Working	Answer
simplifying ratios		
3.1	$\text{HCF} = 5$ $15 \div 5 = 3,\ 25 \div 5 = 5$	**3 : 5**
3.2	$\text{HCF} = 14$ $28 \div 14 = 2,\ 42 \div 14 = 3$	**2 : 3**

3.3	$\text{HCF} = 12$ $36 \div 12 = 3,\ 48 \div 12 = 4$	**3 : 4**
3.4	$\text{HCF} = 15$ $30 \div 15 = 2,\ 45 \div 15 = 3$	**2 : 3**
3.5	$\text{HCF} = 8$ $16 \div 8 = 2,\ 40 \div 8 = 5$	**2 : 5**
3.6	$\text{HCF} = 9$ $27 \div 9 = 3,\ 63 \div 9 = 7$	**3 : 7**
sharing in a ratio		
3.7	$2 + 3 = 5$ $60 \div 5 = 12$ $2 \times 12 = 24$ $3 \times 12 = 36$	**Omar 24 KD, Mona 36 KD**
3.8	$5 + 3 = 8$ $72 \div 8 = 9$ $5 \times 9 = 45$ $3 \times 9 = 27$	**Fahad 45 KD, Laila 27 KD**
3.9	$7 + 2 = 9$ $90 \div 9 = 10$ $7 \times 10 = 70$ $2 \times 10 = 20$	**Noor 70 KD, Ahmad 20 KD**
3.10	$3 + 5 = 8$ $40 \div 8 = 5$ $3 \times 5 = 15$ $5 \times 5 = 25$	**Yousef 15 KD, Huda 25 KD**
3.11	$5 + 7 = 12$ $84 \div 12 = 7$ $5 \times 7 = 35$ $7 \times 7 = 49$	**Sara 35 KD, Ali 49 KD**

3.12	$3 + 7 = 10$ $100 \div 10 = 10$ $3 \times 10 = 30$ $7 \times 10 = 70$	**Maha 30 KD, Faisal 70 KD**
recipes		
3.13	$240 \div 3 = 80$ $2 \times 80 = 160$	**160 g**
3.14	$420 \div 7 = 60$ $2 \times 60 = 120$	**120 g**
3.15	$150 \div 2 = 75$ $5 \times 75 = 375$	**375 ml**
3.16	$360 \div 4 = 90$ $1 \times 90 = 90$	**90 g**
3.17	$120 \div 3 = 40$ $5 \times 40 = 200$	**200 g**
3.18	$450 \div 5 = 90$ $4 \times 90 = 360$	**360 g**
finding the total		
3.19	$18 \div 3 = 6$ $8 \times 6 = 48$	**48**
3.20	$28 \div 4 = 7$ $11 \times 7 = 77$	**77**
3.21	$12 \div 2 = 6$ $11 \times 6 = 66$	**66**
3.22	$35 \div 5 = 7$ $8 \times 7 = 56$	**56**
3.23	$27 \div 3 = 9$ $11 \times 9 = 99$	**99**

3.24	$32 \div 4 = 8$ $9 \times 8 = 72$	**72**

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Q	Working	Answer
substitution into P = 2l + 2w		
4.1	$P = 2 \times 8 + 2 \times 5$ $= 16 + 10$ $= 26$	$P = 26$
4.2	$P = 2 \times 12 + 2 \times 7$ $= 24 + 14$ $= 38$	$P = 38$
4.3	$P = 2 \times 3.5 + 2 \times 2$ $= 7 + 4$ $= 11$	$P = 11$
4.4	$P = 2 \times 10 + 2 \times 4.5$ $= 20 + 9$ $= 29$	$P = 29$
4.5	$P = 2 \times 15 + 2 \times 9$ $= 30 + 18$ $= 48$	$P = 48$
4.6	$P = 2 \times 6 + 2 \times 2.5$ $= 12 + 5$ $= 17$	$P = 17$
substitution into v = u + at		
4.7	$v = 10 + 4 \times 3$ $= 10 + 12$ $= 22$	$v = 22$
4.8	$v = 5 + 6 \times 7$ $= 5 + 42$ $= 47$	$v = 47$

4.9	$v = 15 + 2 \times 9$ $= 15 + 18$ $= 33$	$v = 33$
4.10	$v = 3 + 5 \times 8$ $= 3 + 40$ $= 43$	$v = 43$
4.11	$v = 20 + 3 \times 4$ $= 20 + 12$ $= 32$	$v = 32$
4.12	$v = 7 + 9 \times 2$ $= 7 + 18$ $= 25$	$v = 25$
using a formula		
4.13	$35n + 15 = 155$ $35n = 140$ $n = 4$	**4 hours**
4.14	$25n + 10 = 135$ $25n = 125$ $n = 5$	**5 hours**
4.15	$50n + 30 = 280$ $50n = 250$ $n = 5$	**5 hours**
4.16	$45n + 20 = 200$ $45n = 180$ $n = 4$	**4 hours**
4.17	$60n + 40 = 340$ $60n = 300$ $n = 5$	**5 hours**
4.18	$20n + 15 = 135$ $20n = 120$ $n = 6$	**6 hours**

rearranging formulae		
4.19	**(a)** $y-2=5x$ $x=\dfrac{y-2}{5}$ **(b)** $w=\dfrac{A}{l}$	**(a)** $x=\dfrac{y-2}{5}$ **(b)** $w=\dfrac{A}{l}$
4.20	**(a)** $y+7=2x$ $x=\dfrac{y+7}{2}$ **(b)** $h=\dfrac{V}{A}$	**(a)** $x=\dfrac{y+7}{2}$ **(b)** $h=\dfrac{V}{A}$
4.21	**(a)** $y-1=6x$ $x=\dfrac{y-1}{6}$ **(b)** $d=\dfrac{C}{\pi}$	**(a)** $x=\dfrac{y-1}{6}$ **(b)** $d=\dfrac{C}{\pi}$
4.22	**(a)** $y+4=3x$ $x=\dfrac{y+4}{3}$ **(b)** $s=\dfrac{P}{4}$	**(a)** $x=\dfrac{y+4}{3}$ **(b)** $s=\dfrac{P}{4}$
4.23	**(a)** $y-3=8x$ $x=\dfrac{y-3}{8}$ **(b)** $a=\dfrac{F}{m}$	**(a)** $x=\dfrac{y-3}{8}$ **(b)** $a=\dfrac{F}{m}$
4.24	**(a)** $y+9=4x$ $x=\dfrac{y+9}{4}$ **(b)** $t=\dfrac{D}{s}$	**(a)** $x=\dfrac{y+9}{4}$ **(b)** $t=\dfrac{D}{s}$